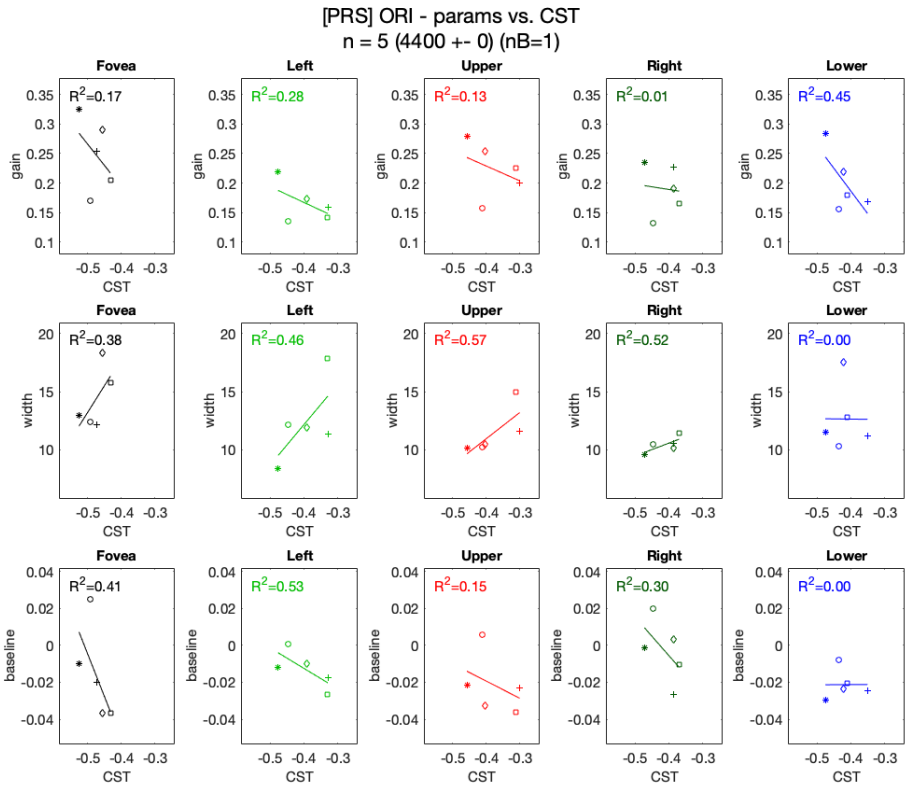
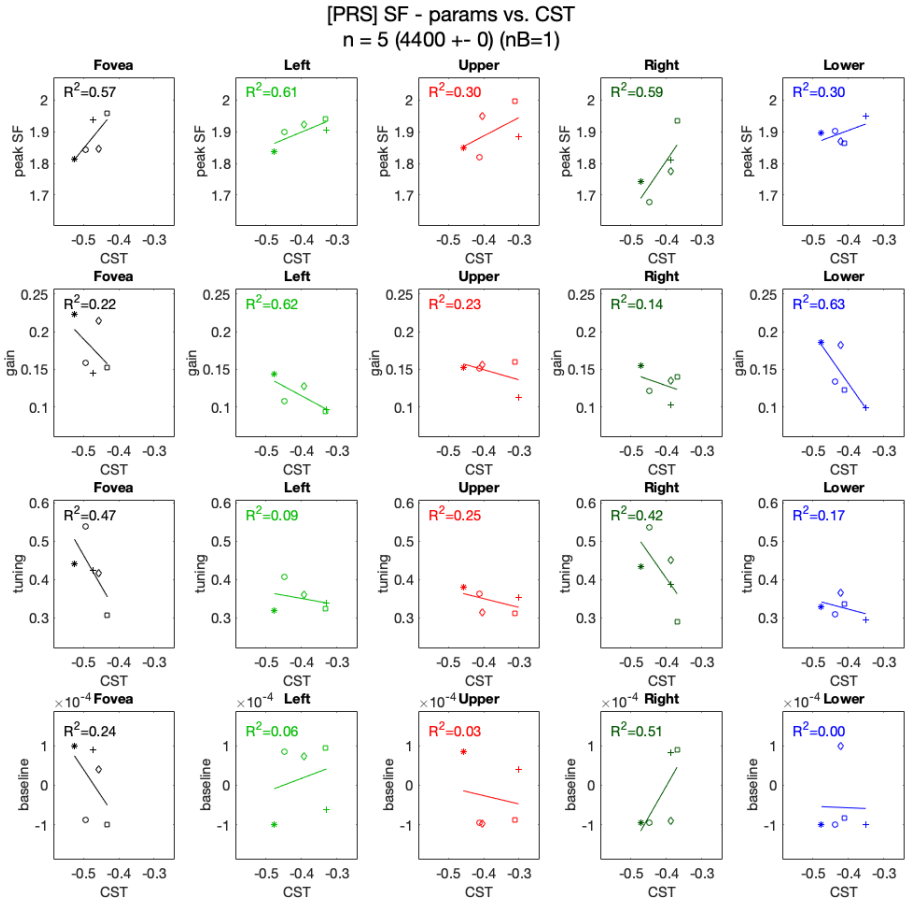
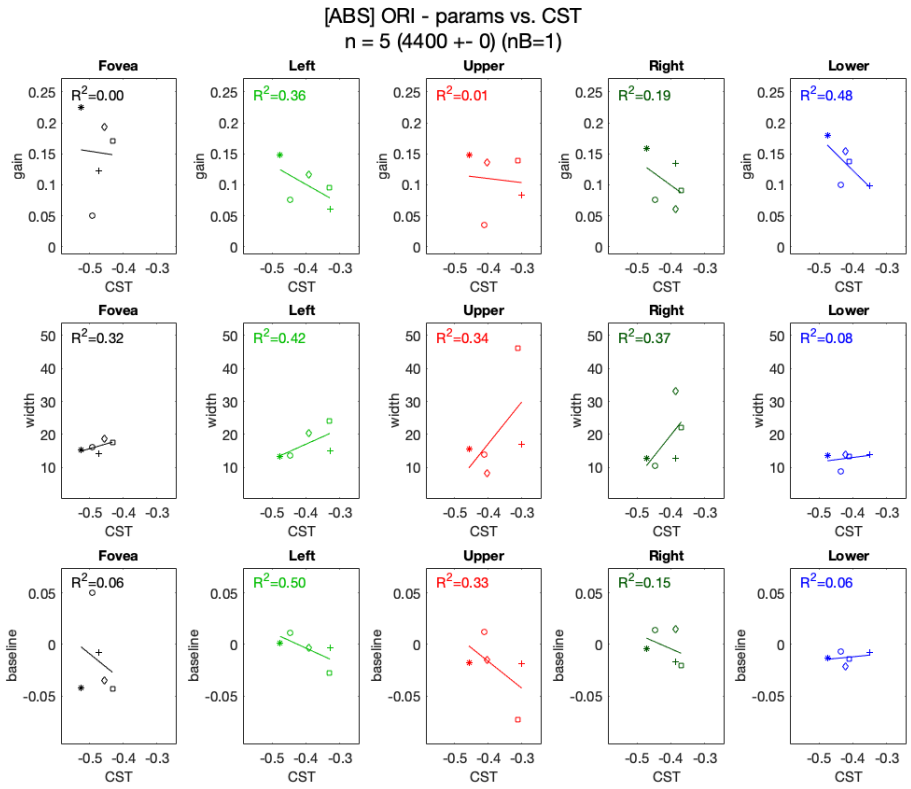
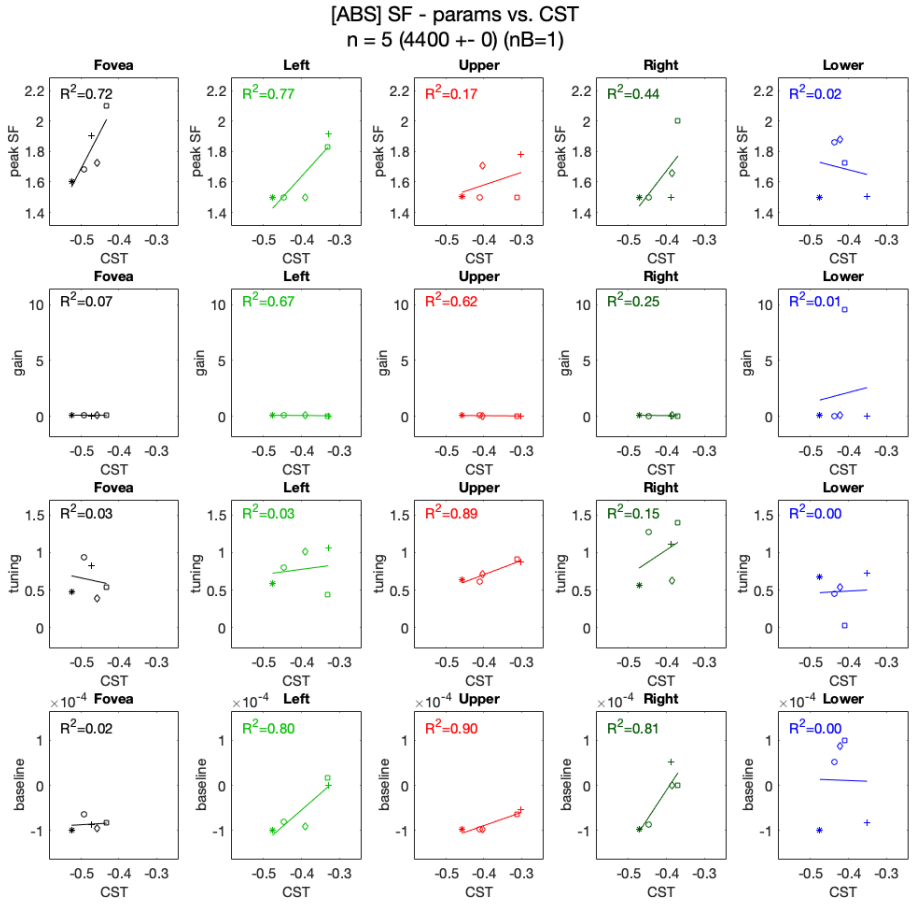
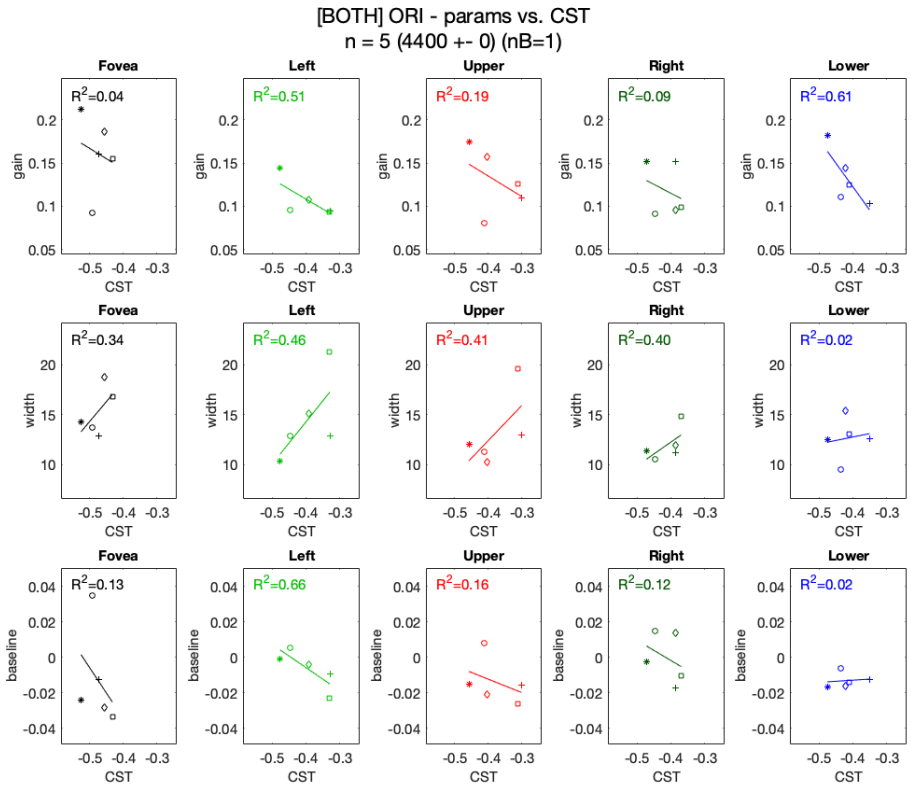
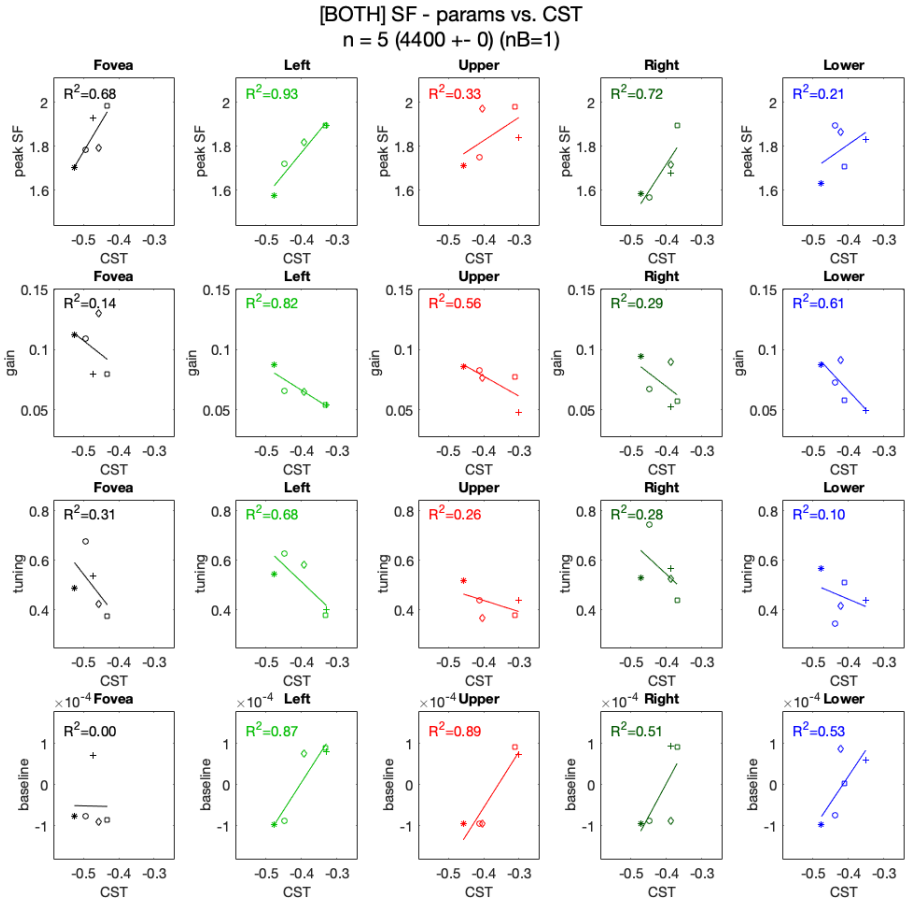

[PRS] ORI - params vs. CST
n = 5 (4400 +- 0) (nB=1)
Fovea
Left
Upper
Right
Lower
gain
width
baseline
CST
R2=0.17
R2=0.28
R2=0.13
R2=0.01
R2=0.45
R2=0.38
R2=0.46
R2=0.57
R2=0.52
R2=0.00
R2=0.41
R2=0.53
R2=0.15
R2=0.30
R2=0.00

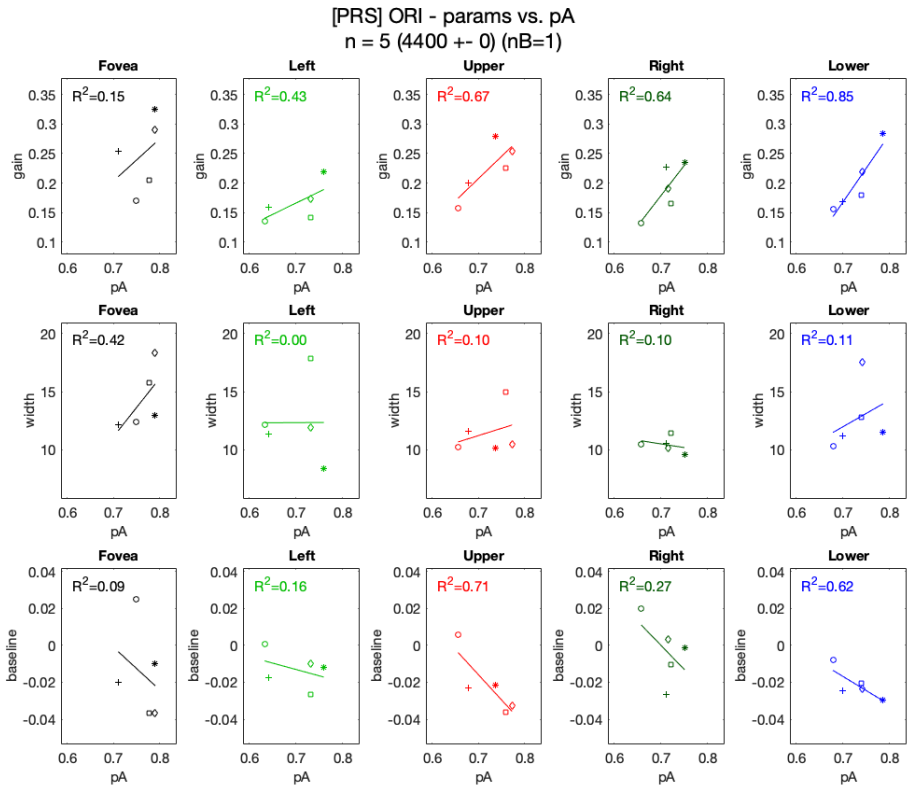

[PRS] SF - params vs. CST
n = 5 (4400 +- 0) (nB=1)
Fovea
Left
Upper
Right
Lower
peak SF
gain
tuning
baseline
CST
R2=0.57
R2=0.61
R2=0.30
R2=0.59
R2=0.30
R2=0.22
R2=0.62
R2=0.23
R2=0.14
R2=0.63
R2=0.47
R2=0.09
R2=0.25
R2=0.42
R2=0.17
R2=0.24
R2=0.06
R2=0.03
R2=0.51
R2=0.00

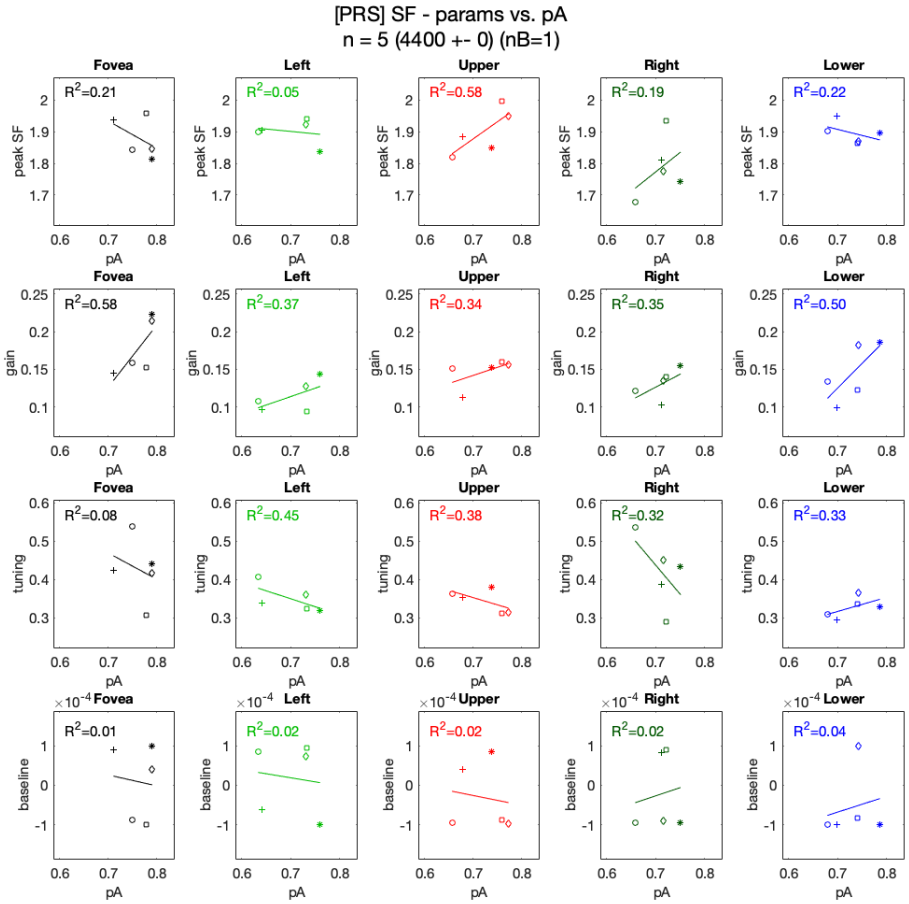

[ABS] ORI - params vs. CST
n = 5 (4400 +- 0) (nB=1)
Fovea
Left
Upper
Right
Lower
R2=0.00
R2=0.36
R2=0.01
R2=0.19
R2=0.48
R2=0.32
R2=0.42
R2=0.34
R2=0.37
R2=0.08
R2=0.06
R2=0.50
R2=0.33
R2=0.15
R2=0.06
gain
width
baseline
CST

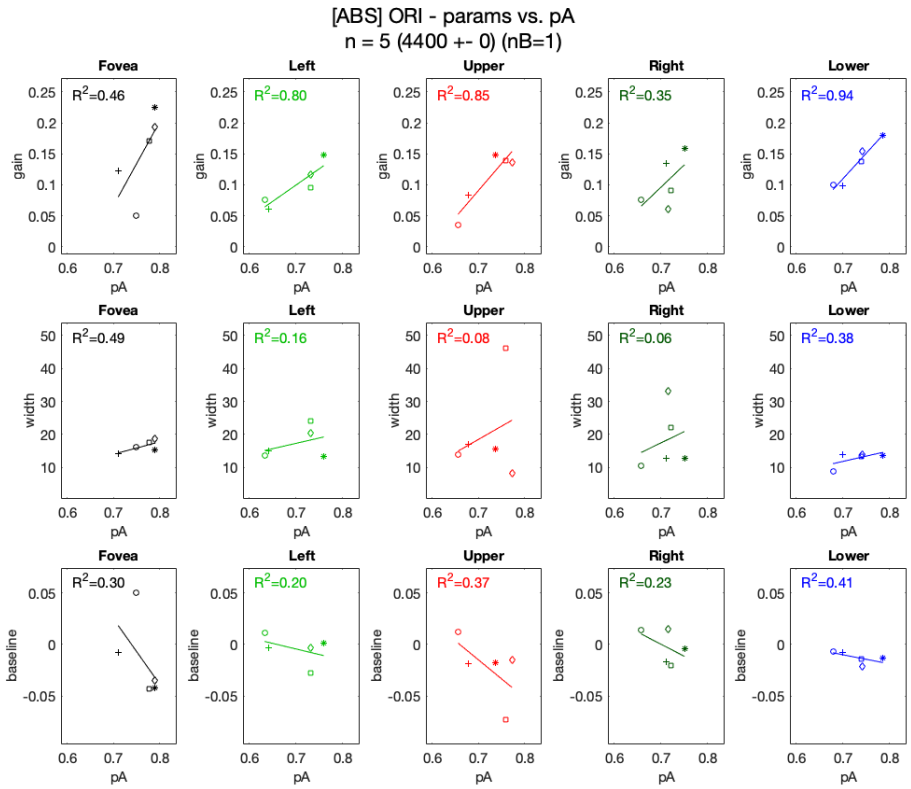

[ABS] SF - params vs. CST
n = 5 (4400 +- 0) (nB=1)
Fovea
Left
Upper
Right
Lower
peak SF
gain
tuning
baseline
CST
R2=0.72
R2=0.77
R2=0.17
R2=0.44
R2=0.02
R2=0.07
R2=0.67
R2=0.62
R2=0.25
R2=0.01
R2=0.03
R2=0.03
R2=0.89
R2=0.15
R2=0.00
R2=0.02
R2=0.80
R2=0.90
R2=0.81
R2=0.00

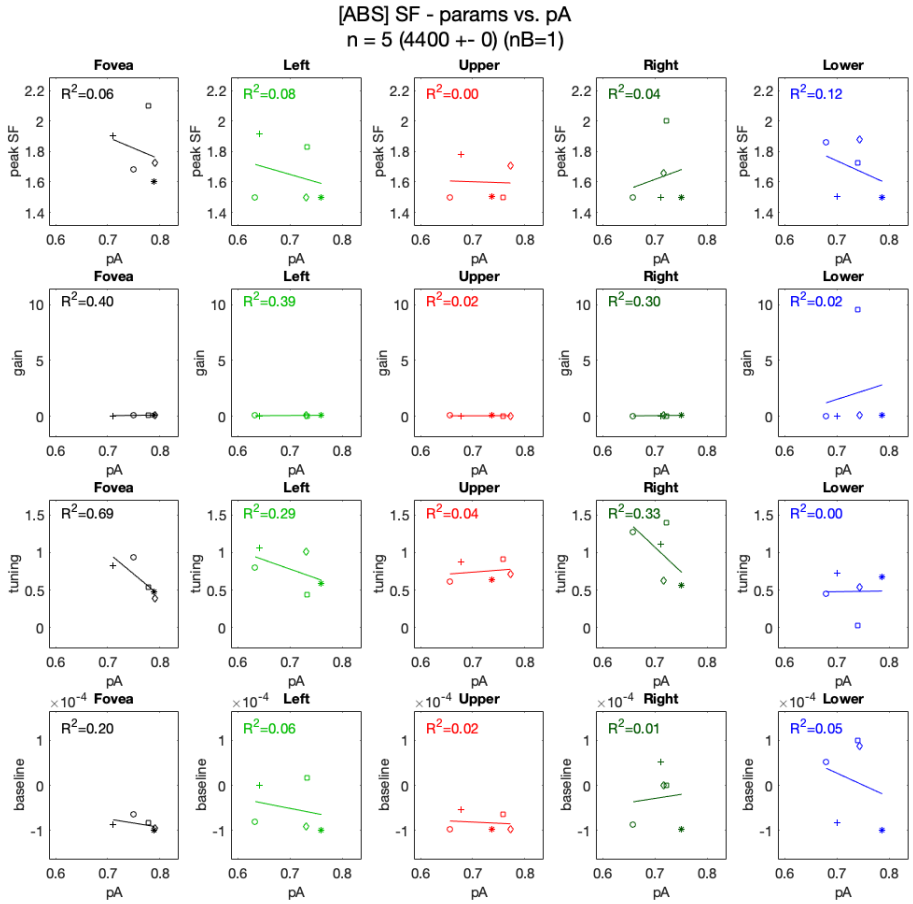

[BOTH] ORI - params vs. CST
n = 5 (4400 +- 0) (nB=1)
Fovea
Left
Upper
Right
Lower
R2=0.04
R2=0.51
R2=0.19
R2=0.09
R2=0.61
R2=0.34
R2=0.46
R2=0.41
R2=0.40
R2=0.02
R2=0.13
R2=0.66
R2=0.16
R2=0.12
R2=0.02
gain
width
baseline
CST
-0.5 -0.4 -0.3

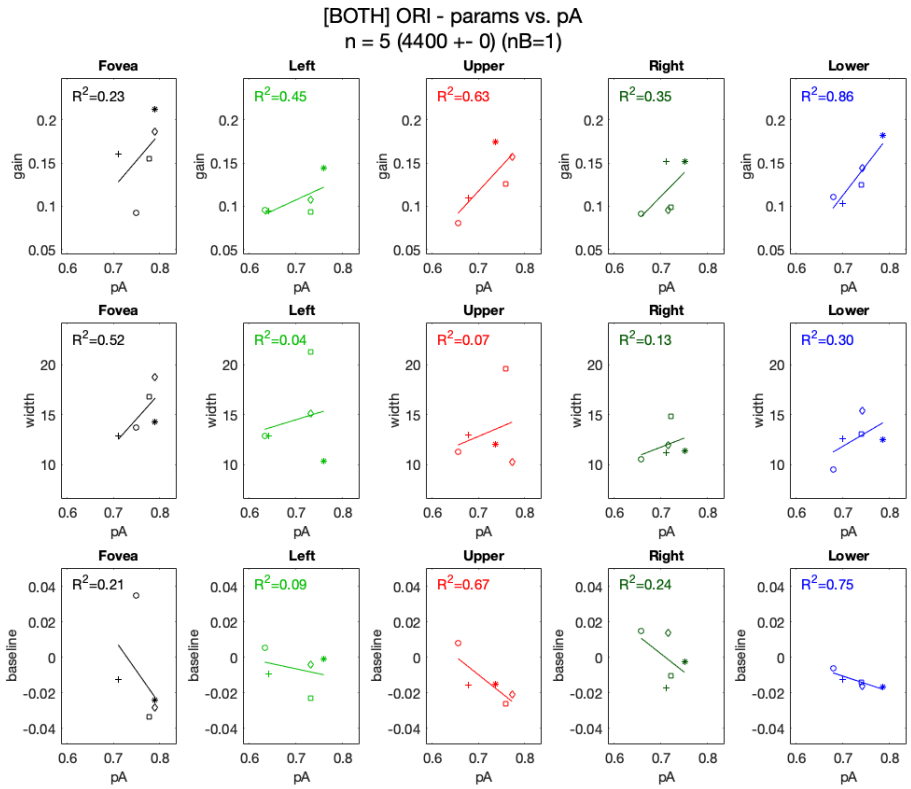

[BOTH] SF - params vs. CST
n = 5 (4400 +- 0) (nB=1)
Fovea
Left
Upper
Right
Lower
peak SF
gain
tuning
baseline
CST
R2=0.68
R2=0.93
R2=0.33
R2=0.72
R2=0.21
R2=0.14
R2=0.82
R2=0.56
R2=0.29
R2=0.61
R2=0.31
R2=0.68
R2=0.26
R2=0.28
R2=0.10
R2=0.00
R2=0.87
R2=0.89
R2=0.51
R2=0.53

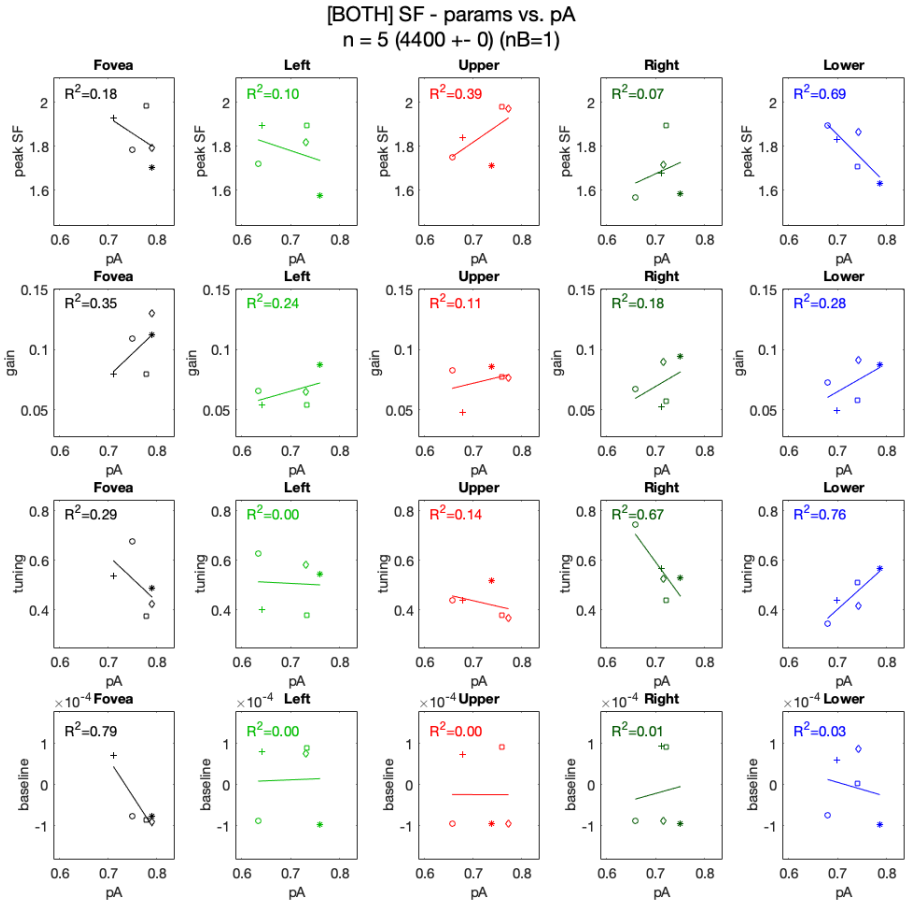

[PRS] ORI - params vs. pA
n = 5 (4400 +- 0) (nB=1)
Fovea
Left
Upper
Right
Lower
gain
width
baseline
pA
R2=0.15
R2=0.43
R2=0.67
R2=0.64
R2=0.85
R2=0.42
R2=0.00
R2=0.10
R2=0.10
R2=0.11
R2=0.09
R2=0.16
R2=0.71
R2=0.27
R2=0.62


[PRS] SF - params vs. pA
n = 5 (4400 +- 0) (nB=1)
Fovea
Left
Upper
Right
Lower
R2=0.21
R2=0.05
R2=0.58
R2=0.19
R2=0.22
peak SF
R2=0.58
R2=0.37
R2=0.34
R2=0.35
R2=0.50
gain
R2=0.08
R2=0.45
R2=0.38
R2=0.32
R2=0.33
tuning
R2=0.01
R2=0.02
R2=0.02
R2=0.02
R2=0.04
baseline
pA


[ABS] ORI - params vs. pA
n = 5 (4400 +- 0) (nB=1)
Fovea
Left
Upper
Right
Lower
R2=0.46
R2=0.80
R2=0.85
R2=0.35
R2=0.94
R2=0.49
R2=0.16
R2=0.08
R2=0.06
R2=0.38
R2=0.30
R2=0.20
R2=0.37
R2=0.23
R2=0.41
gain
width
baseline
pA


[ABS] SF - params vs. pA
n = 5 (4400 +- 0) (nB=1)
Fovea
Left
Upper
Right
Lower
peak SF
gain
tuning
baseline
pA
R2=0.06
R2=0.08
R2=0.00
R2=0.04
R2=0.12
R2=0.40
R2=0.39
R2=0.02
R2=0.30
R2=0.02
R2=0.69
R2=0.29
R2=0.04
R2=0.33
R2=0.00
R2=0.20
R2=0.06
R2=0.02
R2=0.01
R2=0.05


[BOTH] ORI - params vs. pA
n = 5 (4400 +- 0) (nB=1)
Fovea
Left
Upper
Right
Lower
R2=0.23
R2=0.45
R2=0.63
R2=0.35
R2=0.86
R2=0.52
R2=0.04
R2=0.07
R2=0.13
R2=0.30
R2=0.21
R2=0.09
R2=0.67
R2=0.24
R2=0.75
gain
width
baseline
pA



thresh_final_allSubj_log = log10(thresh_final_allSubj); thresh_min_log = min(thresh_final_allSubj_log(:)); thresh_max_log = max(thresh_final_allSubj_log(:));

%% 3 comparisons % get idvd params of each (combined) location

params_allComp_allSubj = nan(6, 2, nsubj, ntypes, 3);

compInd = [1,8,6,7,5,3]; ncomp_ = length(compInd);

for icomp = compInd kernel2D = data2D_allSubj_comp{icomp}; for ikernel = 1:2 switch ikernel case 1 x = filtersOri_all - 90; kernel1D_allSubj = dataOri_allSubj_comp{icomp}; % nsubj x ntypes x icomp * nfilters model = 1; case 2 x = filtersSF_all_cut; kernel1D_allSubj = dataSF_allSubj_comp{icomp}; model = 2; end

```
            for isubj = 1:nsubj
                for itype = 1:ntypes
                    kernel1D = squeeze(kernel1D_allSubj(isubj, itype, :));
                    [~, params_est, ~, ~] = SX_sim08_fit(x, kernel1D', model, f
                    params_allComp_allSubj(icomp, ikernel, isubj, itype, :) = p

                end
            end
        end
```

```
end
```

%% ylimit{1,1} = [0, .45]; ylimit{1,2} = [0, 50]; ylimit{1,3} = [-0.2, .05]; ylimit{2,1} = [0,1]; ylimit{2,2} = [0.05, 2.35]; ylimit{2,3} = [0, 2.5];

for ikernel = 1:2 switch ikernel case 1 nparams = nparamsOri; namesModelParams = namesModelParamsOri; sgtitle_= 'Orientation'; case 2 nparams = nparamsSF-1; namesModelParams = namesModelParamsSF(1:nparams); sgtitle_ = 'SF'; end

```
figure('Position', [0 500 1500 nparams*300])
for icomp = [1,8,6,7,5,3] % fovea, peri, HM, VM, lower VM, upper VM
    switch icomp
        case 1, thresh = thresh_final_allSubj_log(:, 1);ind = 1;
        case 2, thresh = mean(thresh_final_allSubj_log(:, 2:5),2); ind
        case 3, thresh = mean(thresh_final_allSubj_log(:, [2,4]),2); in
        case 4, thresh = mean(thresh_final_allSubj_log(:, [3,5]),2); in
        case 5, thresh = mean(thresh_final_allSubj_log(:, 5),2);ind = 1
        case 6, thresh = mean(thresh_final_allSubj_log(:, 3),2);ind = 2
    end

    for iparam = 1:nparams
        pp = squeeze(params_allComp_allSubj(icomp, ikernel, :, :, ipara
        if ikernel == 2 && iparam == 1 % SF peak
            pp = log2(pp);
        end

        subplot(nparams, ncomp_, (iparam-1)*ncomp_ + find(icomp==compIn
        Rsquared = nan(1,ntypes);
        for itype = 1:ntypes
            % plot data
            for isubj = 1:nsubj
                plot(thresh(isubj), pp(isubj,itype), [marks_allSubj{isu
            end

            % corr analysis
            y = pp(:, itype);
            [r, p] = corr(thresh, y);
            if p < .05, linestyle = '-'; else, linestyle = '--'; end

            % linear regression
            lm_param = polyfit(thresh, y, 1);
            y_pred = polyval(lm_param, thresh);
            plot(thresh, y_pred, [linestyle, colorsType{itype}], 'handl
            Rsquared(itype) = 1-sumsqr(y-y_pred)/sumsqr(y-mean(y));
            if ikernel == 2 && iparam == 1, textRsquare_y = log2(2.6)-i
            else, textRsquare_y = max(pp(:)) - itype*std(pp(:))*.3;
            end
            text(min(thresh) - .18, textRsquare_y, sprintf('R^2 = %.3f'

        end
        %             Rsquared
        xlim([min(thresh) - .2, max(thresh) + .1])
        xticks(-.6:.1:-.2)
        xticklabels(round(10.^(-.6:.1:-.2), 2))
        %             ylim(ylimit{ikernel, iparam})
```

```
                %             yticks(ylimit{ikernel, iparam})

                %             xlim([thresh_min_log - .05, thresh_max_log + .01]
                if iparam + icomp == 2
                    legend(subjList, 'location', 'best', 'orientation', 'horizo
                    xlabel('cst threshold')
                end
%               if icomp == 1, ylabel(namesModelParams{iparam}), end
%               if iparam == 1, title(locNames_comp{icomp}), end
                if ikernel == 2 && iparam == 1 % SF peak
                    yline(1, 'handlevisibility', 'off');
                    ylim([log2(1.4), log2(2.6)]),
                    yticks([log2(1.4), 1, log2(2.6)]),
                    yticklabels([1.4, 2, 2.6])
                end
            end
        end

            set(findall(gcf, '-property', 'FontSize'), 'FontSize',13)
        sgtitle(sgtitle_, 'FontSize',20)
    end
```

%% 5 locations for ikernel = 1:2 switch ikernel case 1 nparams = nparamsOri; params_allSubj = paramsOri_allSubj; namesModelParams = namesModelParamsOri; sgtitle_ = 'Orientation'; case 2 nparams = nparamsSF-1; params_allSubj = paramsSF_allSubj; namesModelParams = namesModelParamsSF(1:n-params); sgtitle_ = 'SF'; end

```
        figure('Position', [0 500 1500 nparams*250])
        for iLoc = 1:nLoc
            thresh = thresh_final_allSubj_log(:, iLoc);
            for iparam = 1:nparams
                subplot(nparams, nLoc, (iparam-1)*nLoc + iLoc), hold on
                titles = cell(1,ntypes);
                for itype = 1:ntypes

                    params_allComp_allSubj = params_allSubj(:, itype, iLoc, ipa

                    if ikernel == 2 && iparam == 1, params_allComp_allSubj = lo

                    % plot data
                    for isubj = 1:nsubj
                        plot(thresh(isubj), params_allComp_allSubj(isubj), [mar
                    end
                    if iparam + iLoc == 2, legend(subjList, 'location', 'best',

                    % corr analysis
                    [r, p] = corr(thresh, params_allComp_allSubj);

                    % linear regression
                    lm_param = polyfit(thresh, params_allComp_allSubj, 1);
                    thresh_ = linspace(min(thresh), max(thresh), 30);
                    params_pred = polyval(lm_param, thresh_);
                    if p < .05, linestyle = '-'; else, linestyle = '--'; end
                    plot(thresh_, params_pred, [linestyle, colorsType{itype}],

                    titles{itype} = sprintf('r = %.2f, p = %.2f', r, p);
```

```
                end
                xlim([min(thresh) - .1, max(thresh) + .1])
                xticks(-.6:.1:-.2)
                xticklabels(round(10.^(-.6:.1:-.2), 2))
                %             xlim([thresh_min_log - .05, thresh_max_log + .01]
                if iparam + iLoc == 2, xlabel('cst threshold'), end
                if iLoc == 1, ylabel(namesModelParams{iparam}), end
                if iparam == 1, title(namesLoc2D{iLoc}), end
                if ikernel == 2 && iparam == 1
                    yline(1, 'handlevisibility', 'off');
                    ylim([log2(1.4), log2(2.2)]),
                    yticks([log2(1.4), 1, log2(2.2)]),
                    yticklabels([1.4, 2, 2.2])
                end
                %             title(sprintf('%s, %s\n%s\n%s\n%s', locNames{iLoc
            end
        end

        set(findall(gcf, '-property', 'FontSize'), 'FontSize',15)
        sgtitle(sgtitle_, 'FontSize',20)
    end
```

Published with MATLAB® R2019a